

SYNOPSIS

At about 2000 on 9 April 2020, the UK registered dredger *Shearwater* was immobilised after its propeller shafts were fouled by a towline being used to tow the barge *Agem One*. The dredger and barge collided with each other repeatedly resulting in *Shearwater* being holed and flooded, before the towline parted and *Agem One* drifted away. There was no pollution or injury.

Shearwater had been towing *Agem One* in an alongside configuration on a coastal passage when a significant swell was encountered. This made the alongside tow untenable, causing *Shearwater*'s crew to switch to an astern tow. Within minutes of switching, the 80m towline failed. Shortly after reconnecting the towline, it failed again, and the decision was made to abort the planned passage and seek shelter at Kinlochbervie. During the passage to Kinlochbervie the crew had reverted to an alongside tow and, in preparation for entering the narrow channel into the harbour, the towing arrangement was again reconfigured to tow the barge astern. It was during this evolution that the towline became fouled around *Shearwater*'s propeller shafts and the immobilised dredger was damaged. The situation was eventually brought under control after the intervention of a lifeboat, the emergency towing vessel, *Ievoli Black*, and the workboat *Forth Drummer*.

The accident happened because there was insufficient planning, risk assessments or safe systems of work for the towing operation being conducted. *Shearwater* was not suitable for use as a coastal towing vessel especially through hazardous areas such as the Pentland Firth, and the crew did not have the necessary competence to undertake the operation.

Shearwater was too large for certification as a small commercial vessel but under the tonnage requiring a safe manning certificate or safety management system. This investigation has identified that the flag state's arrangements for certifying *Shearwater* using exemptions from the Load Line Regulations did not provide sufficient guidance to assure safe operation of the vessel.

Since the accident, *Shearwater*'s owner has purchased a small tug for use when repositioning barges. Nevertheless, recommendations have been made to *Shearwater*'s owner to assess all on board hazards and provide safe systems of work to mitigate the foreseeable risks, and to ensure the vessel is safely manned. This report also makes a recommendation to the Maritime and Coastguard Agency to ensure that certification of vessels such as *Shearwater* includes the application of all appropriate regulatory conditions relevant to the vessel's intended function and area of operations.

1.2 BACKGROUND

On 25 March 2020, the grab hopper dredger *Shearwater*, towing the barge *Agem One*, departed from Glasgow, Scotland, intending to transit to Eyemouth, Scotland in a series of short passages (**Figure 1**). The purpose of the voyage was to reposition for the start of a new dredging contract. There were four persons on board *Shearwater*: the owner², the master, an able-bodied seaman (AB) and an excavator driver. *Agem One* was laden with a 45 tonne (t) excavator. Prior to the accident, the dredger and barge stopped at Campbeltown, Oban, Lochaline (where a short period of dredging operations was conducted), Kyle of Lochalsh and Gairloch. At 0800 on 8 April 2020, *Shearwater* and *Agem One* left Gairloch and headed north towards Cape Wrath via a short stopover near Stoerhead (**Figure 1**).

1.3 NARRATIVE

1.3.1 Events leading up to the collision

Early in the morning on 9 April 2020, *Shearwater* departed from Stoerhead for passage to Loch Eriboll (**Figure 2**). *Shearwater* was towing *Agem One* alongside (on the dredger's port side) and was proceeding at a speed of between 3 and 5.2 knots (kts). The master was on the bridge and in control of the vessel, and *Shearwater*'s owner was overseeing towing operations. *Shearwater* passed Cape Wrath at about 1030 (**Figure 2**) and started to encounter a long swell with a height of approximately 2.5 metres (m), causing *Agem One* to move heavily against the dredger's side. The alongside towing configuration was judged untenable by *Shearwater*'s owner who ordered that *Agem One* be moved to an astern tow.

The AB and the excavator driver stepped across to *Agem One* from *Shearwater*'s deck and prepared for an astern tow by connecting the shackle on the barge's bridle to the towline's hard thimble eye. Both crewmen then returned to *Shearwater*, *Agem One* was cast off and the 80 metre (m) towline payed out. After *Shearwater* had towed *Agem One* a short distance, the towline broke and the barge drifted free.

Shearwater's master then manoeuvred to retrieve *Agem One* and the two crewmen once again stepped across to the barge to reattach the tow. The towline had failed at the point of connection to the barge's bridle and the hard eye was observed to be distorted, so the crewmen tied a bowline knot in the towline as an alternative connection. The passage towards Loch Eriboll continued but the towline failed again with *Agem One* needing to be retrieved and the tow reattached once more.

At about 1200, following the repeated failures of the towline, the owner and master decided to abort the passage and head back towards shelter in Kinlochbervie.

1.3.2 Immobilisation, collisions, and loss of the barge

At about 1900, *Shearwater* was just over 3 nautical miles (nm) west of Kinlochbervie (**Figure 2**) with *Agem One* back in an alongside towing arrangement. In this area, there was a moderate breeze, a short, choppy sea and good visibility. A decision was then taken to revert to an astern tow due to the narrow width of navigable water available in the approaches to Kinlochbervie.

² The sole director of *Shearwater*'s registered owning company, Northern Dredging Limited, was on board and is referred to as the 'owner' in this report.

Agem One was cast off with the excavator driver on board and the owner and the AB passed the towline from *Shearwater's* aft deck down to him to connect up. Once connected, the master needed to go astern to close the gap between the dredger and the barge so that the excavator driver could be retrieved back on board.

Prior to applying astern propulsion, the owner, who was on the bridge, asked the AB to confirm that the towline was clear of the water. The AB shouted back that it was, so the master applied astern power on both shafts. When *Shearwater* was adjacent to the barge, the excavator driver scrambled back across on to the dredger's side deck; at the same time, both engines stopped. The master restarted the engines and attempted to engage propulsion but neither engine would clutch in.

The excavator driver then went to the engine room to assess the situation and found that the port shaft was dislodged from the gearbox and the gearbox's casing was damaged. He went to the bridge and briefed the master and the owner about the damaged propulsion. By this time, the owner and master had seen from the aft deck that the shafts were fouled by the towline.

The master was aware that both vessels were then drifting north with the tidal stream at about 1kt and the only safe course of action was to drop the anchor. The port anchor was let go and, as it started to hold, *Shearwater* swung round into the tidal stream; at this point, *Agem One*, which was not under control, made several heavy, damaging collisions with *Shearwater's* port side. The crew tried to regain control of *Agem One* by securing it alongside *Shearwater* but were unable to do so. *Agem One* was then held briefly astern of the dredger by the fouled towline before drifting free when the towline broke.

When the crew checked *Shearwater* for damage, water was found entering the port side buoyancy space through damage holes in the shell plating (**Figure 3**).

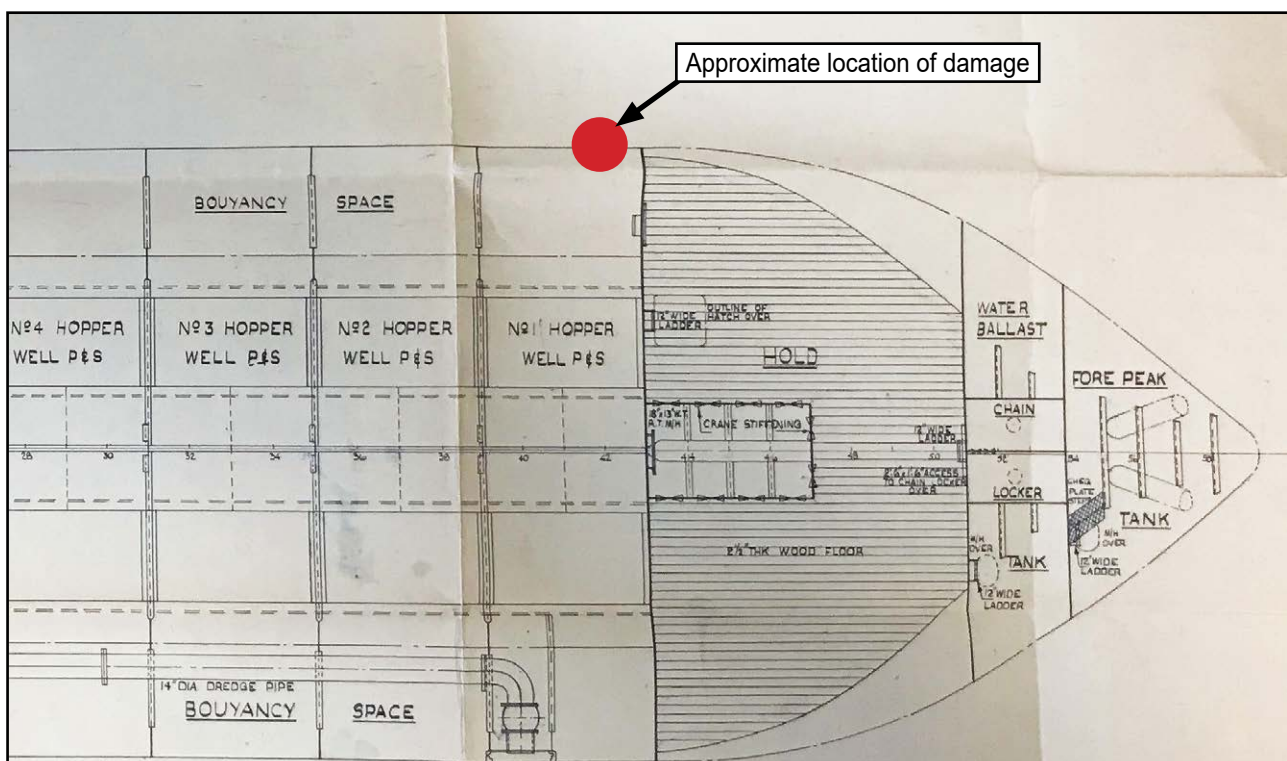


Figure 3: Extract from *Shearwater's* general arrangement showing the approximate location of damage

1.3.3 Actions after *Agem One* broke free

At 2021³, the owner called the coastguard to report that *Agem One*, which was unlit, had broken away from *Shearwater* and was drifting free. The owner also reported that *Shearwater* was immobilised, anchored and taking on water. The coastguard requested that the Lochinver RNLI⁴ all-weather lifeboat locate and retrieve *Agem One*, and the emergency towing vessel (ETV) *Ievoli Black* was tasked to proceed to assist.

Shearwater's crew tried to pump water out of the flooded port buoyancy space using the fixed bilge system but were unsuccessful, so rigged a portable electric submersible pump instead. The crew also attempted to stem the water ingress with rags and sealant. At 2204, the lifeboat located the drifting *Agem One* and commenced towing it back towards *Shearwater*. The lifeboat reached *Shearwater* at around 0130 on 10 April (**Figure 4**) and held position nearby awaiting the arrival of the ETV.

Image courtesy of RNLI Lochinver Lifeboat Facebook page

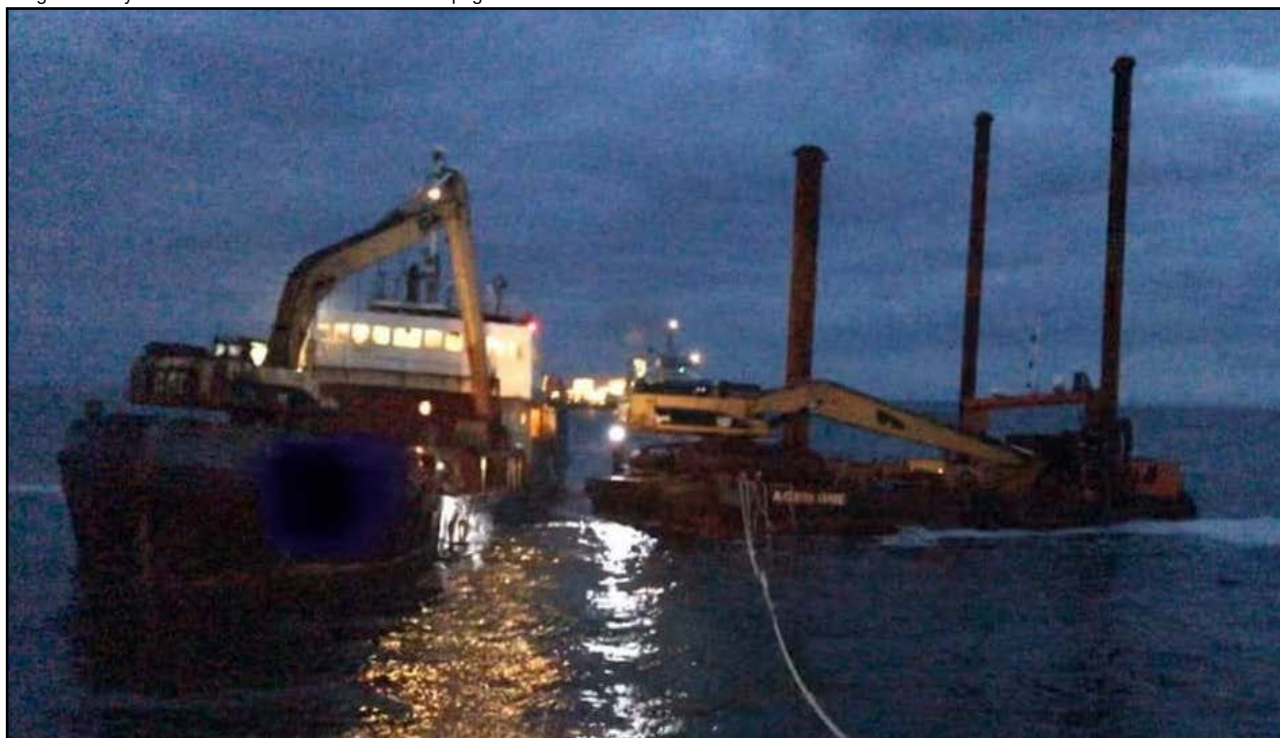


Figure 4: Lochinver lifeboat returning *Agem One* to *Shearwater*

Shearwater's owner informed the coastguard at 0212 that the portable pump was not keeping up with the water ingress and that the dredger had developed a list of between 5 and 10 degrees to port as a result of the flooding. *Ievoli Black* arrived on scene at 0242, then launched a sea boat and transferred two additional portable salvage pumps to *Shearwater*, which were assembled and added to the pumping effort. *Shearwater* was returned to level trim by 0355 and the lifeboat crew prepared to return *Agem One* to the dredger. *Agem One* was secured alongside *Shearwater* at 0547 and the lifeboat headed home; *Ievoli Black* remained standing by to assist.

³ 2021 was also the time of sunset

⁴ Royal National Lifeboat Institution

On board *Shearwater*, the crew rested and waited for the workboat *Forth Drummer*⁵ to arrive. *Agem One* periodically made further heavy contact with *Shearwater* causing more damage to the dredger's hull plating. *Forth Drummer* arrived at 1730 and initially towed *Agem One* into Kinlochbervie. *Forth Drummer* then returned to the anchored *Shearwater* and took the dredger in tow bound for Aberdeen; *Ievoli Black* left the scene at 2200 after this tow was established.

At 0512 on 11 April, when transiting the Pentland Firth, *Forth Drummer* reported that *Shearwater* was again taking on water and had developed a list of around 10 degrees. The tow to Aberdeen was aborted and *Shearwater* was taken into Scrabster (**Figure 5**). Following temporary repairs, the tow was resumed with a second tug, *Shuna*⁶, on 16 April. *Shearwater* entered Aberdeen harbour at around 0600 on 18 April and was taken to dry dock where an inspection found the towline wrapped around both propeller shafts (**Figure 6**), penetration and significant indentation to the port side shell plating and damage to the port shaft and gearbox. *Agem One* was undamaged following the accident and was left at Kinlochbervie anchored by its spud legs; it was eventually retrieved in December 2020.

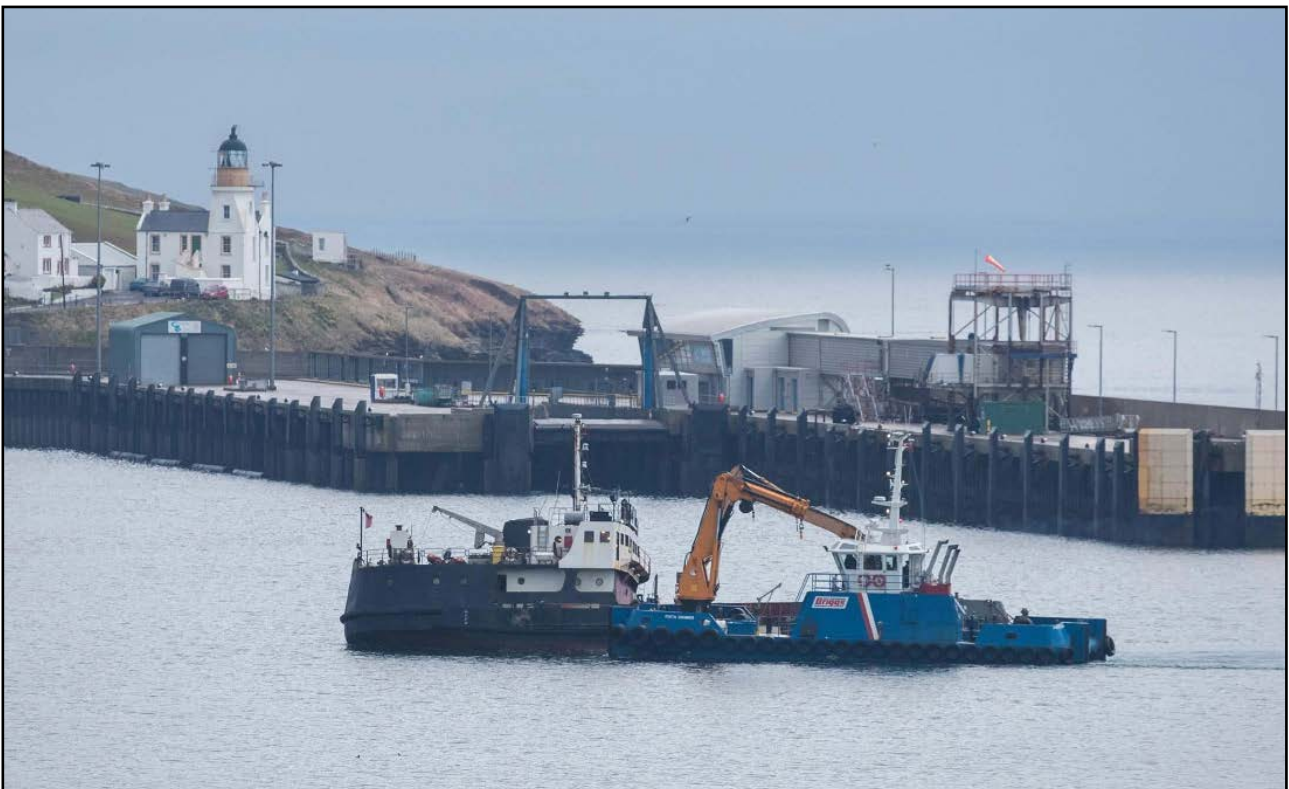


Figure 5: *Shearwater* entering Scrabster harbour being towed alongside *Forth Drummer*

⁵ *Forth Drummer* was a multi-purpose workboat, capable of towing and with a certified bollard pull of 23.5 tonnes

⁶ *Shuna*'s bollard pull was 14 tonnes

Agem One's stability book was approved by SCMS and assessed the barge against the stability criteria in the MCA's Small Commercial Vessel and Pilot Boat Code of Practice (SCV Code)¹⁷. The stability book included a standard stability condition for towing the barge with spud legs raised and carrying an excavator¹⁸.

In 2015 *Agem One* capsized when being towed by a tug. The barge was towed to port in the upturned condition where it was righted and later recommissioned.

1.8 PASSAGE PLANNING

1.8.1 Guidance

SOLAS¹⁹ Chapter V Regulation 34 applied to all vessels on all voyages and required that, prior to proceeding to sea, the master was to ensure that the intended voyage had been planned, taking into account the guidance in the IMO's Resolution A.893(21) *Guidelines for Voyage Planning*. This guidance stated that *'the development of a plan for voyage or passage, as well as the close and continuous monitoring of the vessel's progress and position during the execution of such a plan, are of essential importance for safety of life at sea, safety and efficiency of navigation and protection of the marine environment.'*

The guidance subdivided passage planning into four key stages: appraisal, planning, execution, and monitoring. The initial voyage planning appraisal stage involved the gathering of all information relevant to the intended voyage. The next stage required the detailed planning of the whole voyage from berth to berth. The third and fourth stages were the effective execution of the plan and monitoring the vessel's progress during the implementation phase.

The Pentland Firth (**Figure 1**) is the sea passage between the Scottish mainland and the Orkney Islands. The area is notorious for extreme tidal and sea conditions that must be considered when passage planning. For vessels planning a passage through the Pentland Firth, Admiralty Sailing Directions (North Coast of Scotland Pilot) (NP52) emphasised the strength of the tidal streams and the resultant confused and violent seas. There was a particular hazard for low-powered vessels and the sailing directions stated that *'another factor of safe navigation of Pentland Firth is availability of sufficient power to overcome the strengths of the tidal streams'*.

The admiralty chart of the area²⁰ contained a warning notice about the hazards in the Pentland Firth stating that *'tidal streams, with eddies and turbulence, run strongly through the Pentland Firth. Rates of up to 16kt have been reported...'* A recommendation on navigation also printed on the chart stated that *'mariners intending to use the Pentland Firth should be aware of very strong tidal streams and sets within the area. Difficulties can be encountered when transiting either with or against the tide.'*

¹⁷ The Small Commercial Vessel and Pilot Boat Code of Practice (SCV Code) was published in 2004 by the MCA as an Annex to Marine Guidance Note (MGN) 280(M): *Small Vessels in Commercial Use for Sport or Pleasure, Workboats and Pilot Boats – Alternative Construction Standards*.

¹⁸ *Agem One*'s approved towing stability booklet allowed for an excavator weight up to 51.3t

¹⁹ International Convention on the Safety of Life at Sea, 1974, as amended

²⁰ Chart 1954 – Scotland North Coast – Cape Wrath to Pentland Firth

1.8.2 *Shearwater's* passage plan

Shearwater's primary means of navigation was paper charts. The passage plan on board *Shearwater* consisted of a list of hand written positions with the courses and distances inbetween. There was no information on anticipated navigational hazards, speeds, predicted tidal streams or aids to navigation. The intended route was plotted on a paper chart and replicated on the vessel's electronic charting system (ECS) (**Figure 9**). The master used 4kts as the planning speed for passages towing a barge. Evidence from automatic identification system (AIS) data (**Figures 1 and 2**) indicated that *Shearwater's* passage speeds were between 3kts and 5.2kts.

Shearwater had completed a dredging contract in the river Clyde and the next contracts were in Lochaline, Isle of Mull, then Eyemouth, on the east coast of Scotland. The owner had previously used the Caledonian Canal to reposition *Shearwater* from one side of Scotland to the other (**Figure 1**). At the time of this planned passage, the Caledonian Canal was closed due to the COVID-19 pandemic.

Shearwater's master had not been through the Pentland Firth before although the owner had some prior experience of the area from diving trips. There was a 1968 edition of the Admiralty Tidal Stream Atlas on board *Shearwater*, but there was no copy of NP52 on board.

1.9 TOWING

1.9.1 International guidance

The IMO's circular MSC/Circ.884 of 1998 provided guidelines for the planning, preparation, and standards for safe seagoing towage. This guidance detailed responsibilities regarding towing and stated that the operation should be overseen by a competent towing master with procedures in place to minimise the risk to personnel. The guidance also stated that in preparation for the tow an inspection should be carried out by the tow master. IMO and industry guidance indicated that 500m²¹ was a minimum towline length in benign coastal waters.

1.9.2 Workboat industry guidance

The Workboat Association (WA) guide *The Use of Workboats for Towage* provided industry best practice for towing by workboats²², including towing of unpropelled barges in harbour and at sea.

The WA guidance advised that the tow length needed to be adequate for the sea conditions considering the potential for generation of snatch²³ in the towline. The guidance also advised that '*manufacturer's instructions must be carefully followed, particularly with respect to chafe*' if high modulus polyethylene (HMPE) towlines were used and that all towlines should be monitored and adjusted regularly to prevent chafe. Guidance promoted the fitting of anti-chafe devices such as abrasion

²¹ IMO MSC Circ/884 provided the formula for tow length = bollard pull ÷ minimum breaking load x 1800. This formula also appears in DNVGL-ST-N100 industry guidance which further stated that the minimum length of tow should not be less than 650m in normal weather conditions, and a minimum of 500m in benign coastal weather conditions. The formula to calculate tow length could not be used on board *Shearwater* as the vessel's bollard pull was unknown.

²² Although *Shearwater* was not certified as a workboat, *Agem One* was, making the WA guidance relevant for this nature of tow

²³ Snatch is the generation of large dynamic forces in a towline as it comes taught. Snatch can be countered by the use of long towlines where the catenary acts a damper, or the use of a pennant (or 'stretcher') made of a material with greater elongation characteristics than the main towline and providing additional elasticity.

resistant sleeves or chains in high wear areas. The guidance also detailed the importance of contingency plans for adverse weather, the need to plan for specific requirements at all ports of call, including ports of refuge and the need for risk assessments to cover all aspects of the operation. The WA guidance stated that tow plans containing details of the tow, towing equipment, route for the tow and emergency plans should be prepared by the tow master.

The WA guidance stated that the crew and tow master must be sufficiently experienced and competent for the planned operation. The guidance suggested that competence could be established by reference to the master's experience of towing operations and by holding an MCA towing endorsement. *Shearwater's* master did not hold an MCA towing endorsement.

1.9.3 The tow plan

Shearwater's tow plan (**Annex C**) intended that *Agem One* would be towed alongside when conditions permitted and otherwise in an astern tow arrangement. The weather limits in the tow plan were a maximum of force 4 winds for both tow configurations, and wave heights of less than 1m for alongside towing and 1.75m when towing astern. The plan also included a list of potential ports of refuge and noted that extra care should be taken going through the Pentland Firth. The diagram of the alongside tow described it as a '*hip (side) tow*' and indicated that the tow configuration was for '*categorised waters²⁴ and sheltered conditions*'. For towing astern, the tow plan stated that an 80m kevlar towline would be in use; the associated diagram showed that a polyester towline would be used. The towline in use was HMPE.

The tow plan included a note that the excavator on *Agem One* should be removed at Lochaline unless '*perfect conditions*' were forecast for the remainder of the voyage. When *Shearwater* departed from Lochaline the excavator could not be removed due to local COVID-19 restrictions.

The safety section of the tow plan included the requirements that crew be trained on the snapback and crush zones on *Shearwater* and *Agem One*, and that full personnel protective equipment (PPE) was to be worn on deck. Crew members wore personal flotation devices (PFD) when on deck and on the barge. The tow plan did not include details of emergency towing arrangements or procedures.

The tow plan did not state the communications arrangements during towing operations. Hand-held very high frequency (VHF) radios were carried on board *Shearwater*, although communications during towing evolutions was primarily by voice.

1.9.4 Towing arrangements

During the passage, *Shearwater* towed *Agem One* both alongside and astern. Towing alongside involved attaching the barge directly to the dredger by means of ropes at the bow and stern, and spring lines to take the barge's weight. Where swell was present, there was a risk of interaction between the vessels, so the alongside method was restricted to harbours and calm coastal water towing.

²⁴ Categorised waters refers to inland waterways and UK harbour areas that are not classified by the MCA as the open sea.

On the day of the accident the astern tow was set up according to *Shearwater's* tow plan (**Annex C**). The towline was attached to a port mooring bollard on the aft deck of *Shearwater* and then led through a fairlead. This meant the vessels' separation was about 80m on the day of the accident. At the barge end, the towline was attached to a steel wire rope bridle with a shackle, the bridle was attached to mooring points on either side of the forward end of the barge.

1.9.5 Towline material

Shearwater's towline was an 80m long section of 64 millimetre (mm) diameter HMPE with a hard thimble eye at one end. The manufacturer's datasheet stated that the rope's MBL was 3210 kilonewtons (kN) equating to 327.3t; the rope was buoyant with an elongation of 1%. *Shearwater's* owner purchased the rope in 2019 from a surplus dealer and had used it for towing on one previous occasion.

HMPE rope has many applications in both marine and non-marine sectors and is recognised as having a high strength to weight ratio and low elongation at breaking. The low elongation characteristics of HMPE ropes mean that dynamic loading can introduce shock loads resulting in overload. For this reason, HMPE should be used in combination with a pennant (or 'stretcher') to reduce vulnerability to dynamic snatch loading.

Knots in HMPE ropes can introduce weak points in a towing system and the use of a bowline may reduce the breaking strength of the rope by up to 25%. The parted manufactured eye of *Shearwater's* towline was not recovered from *Agem One* in the aftermath of the accident so the exact failure mechanism could not be determined.

1.9.6 Emergency towing arrangements

Agem One's emergency towline was designed to be rigged such that it could be streamed astern of the barge and picked up in the event of failure of the main towline. The line could then be used to recover the barge in the event of the main towline failing, without having to transfer crew to the barge. *Agem One's* emergency towline was not rigged or streamed on the day of the accident.

1.9.7 Authority to tow

As part of preparation to tow a barge in May 2014, *Shearwater's* owner had approached the MCA to discuss the dredger's suitability for towing. An MCA review of the situation identified the factors that should be considered by the owner prior to using *Shearwater* for towing, including stability, crew competence, tow planning, manoeuvrability, load line limitations and the need to witness a trial tow as *Shearwater* was not a tug. Following this review, the MCA advised the owner that he would need risk assessments and safe systems of work for towing, taking the identified factors into consideration; a load test of *Shearwater's* bollards was also required. The MCA also advised the owner that if towing was to become a regular occurrence, then the terms of the load line exemption certificate would need to be changed. This intention was not followed up by either the owner or the MCA and *Shearwater's* load line exemption certificate that was in force at the time of the accident made no reference to towing.

The owner did commission a naval architecture consultancy to conduct a set of strength calculations on *Shearwater*'s aft deck bollards. The report of these tests concluded that a barge about 20m in length could be towed from the aft deck bollards in calm weather. The results of the strength calculations were communicated to the MCA. *Shearwater* did not have a certified bollard pull value.

1.9.8 MCA voluntary towage endorsement scheme

MGN 468(M) sets out the details of the MCA recognised voluntary towage endorsement scheme. The scheme was developed to help ensure that crew engaged in towage operations have the necessary skills to manage the additional risks of towing operations. The scheme had different endorsement levels to reflect the type of towage to be conducted, including:

- **'General Towage Endorsement'** – *towing and pushing in categorised waters or in limited coastal areas.*
- **'Sea Towage Endorsement'** – *towage of vessels or floating objects at sea'.*

The endorsement required minimum levels of relevant towage sea service, which were 120 days for the 'general' towage endorsement, and a further 180 days with a minimum of 12 separate sea towage operations for the 'sea' towage endorsement.

1.10 RELEVANT PREVIOUS ACCIDENTS

1.10.1 Loss of a crewman overboard from the motor tug *Endurance*

On 5 February 2013, a crewman from the UK registered motor tug *Endurance* fell overboard in rough seas near Beachy Head, England. The crewman fell when attempting to transfer to the unmanned motor cruiser *Sirius M* with a replacement towline after the original towline had parted. Despite efforts by the tug's skipper and a search and rescue operation, the crewman was not recovered, and his body was found some weeks later.

The MAIB's investigation²⁵ identified that the attempt to reconnect a towline between *Endurance* and *Sirius M* was a desperate and ill-considered measure brought about by factors including poor towing practices and a lack of planning, risk assessment and emergency preparedness. The findings included that the risks of the crewman transferring to the unmanned tow had not been appreciated and that the original towline parted due to chafe abrasion because the arrangements did not follow good practice.

1.10.2 Foundering of the dredger *Abigail H*

On 2 November 2008, the 50-year-old grab hopper dredger, *Abigail H*, flooded and foundered when alongside in Heysham. The MAIB's investigation²⁶ found that the cause of the foundering was progressive flooding through a leak in the engine room. The flooding went unnoticed primarily because the regulations that applied to *Abigail H* did not require engine room bilge alarms to be fitted.

²⁵ [MAIB Report 13/2014](#)

²⁶ [MAIB Report 15/2009](#)

Abigail H was over 24m in length and under 500gt and the investigation also observed that the regulations for vessels of this size were complicated and that information from many sources needed to be drawn together to achieve the desired effect of safe operations. The report stated that this contrasted with workboats and fishing vessels, where the regulations, presented as codes of practice, made the requirements clear to owners, and implementation straightforward for surveyors. The report went on to recognise that consolidating the regulations applicable to vessels greater than 24m registered length and less than 500gt into a code of practice would be a substantial task, but that doing so would offer multiple benefits to the MCA and owners.

1.10.3 Capsize and sinking of the cement carrier *Cemfjord*

On 2 January 2015, the Cyprus registered cement carrier *Cemfjord* capsized and foundered in the Pentland Firth with the loss of all eight crew. The MAIB's investigation²⁷ identified that *Cemfjord* was lost in violent sea conditions and the rapid nature of the accident meant that no distress message was issued and there was no time for the crew to abandon ship.

The MAIB's investigation also established that the severe conditions were predictable and commonly experienced in the area.

²⁷ [MAIB Report 8/2016](#)

SECTION 2 – ANALYSIS

2.1 AIM

The aim of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 OVERVIEW

When seeking shelter during a coastal passage towing a laden barge, the dredger *Shearwater* lost propulsion when its propeller shafts were fouled by its own towline; thereafter, control of the situation was lost, and there was nothing the crew could do to prevent heavy and damaging collisions with the barge.

This section of the report will discuss the circumstances of the accident and explain the contributing factors including poor planning, unsafe towing practices and insufficient emergency preparation. *Shearwater's* crew lacked the competence necessary to undertake a towing operation of this nature and the vessel was also unsuitable for the task. The MCA's arrangements for *Shearwater's* certification using exemptions from the Load Line Regulations will also be discussed.

2.3 IMMOBILISATION AND COLLISIONS

Irrespective of the AB's report that the rope was clear of the water and that the rope itself was buoyant, applying astern propulsion when a loose towline was in the water between the dredger's stern and the barge meant that fouling the propeller shafts was almost inevitable. A natural, safe instinct when towing must always be to avoid going astern over your own towline.

Despite the fact that the crew had switched over from an alongside towing configuration to an astern tow previously in the day, there was no safe plan for this evolution and the master's priority, on this occasion, was to recover the stranded crew member back on board *Shearwater* from the barge, hence the need to go astern to bring the vessels together. The fouling of the propellers was coincident with the excavator driver scrambling back on board the dredger.

The owner had returned to the bridge after the towline had been passed down to the excavator driver on the barge. This meant that the evolution on the aft deck was unsupervised. Routinely the owner would check such evolutions, but on this occasion he was not there and did not check the situation with the towline himself. Even if the owner had assessed the situation, the risk of fouling the propellers would still have been very high in the circumstances.

Once both shafts were fouled by an extremely strong rope, there was no prospect of restoring propulsion and this became clear after the excavator driver had witnessed the engine room damage. Thereafter, the only safe course of action was to anchor *Shearwater*. This action may have stabilised the situation for the dredger, but it resulted in the collision and flooding when the heavy, sharp-edged barge was out of control. There was little prospect of the crew regaining control of the barge at this point, and further damaging collisions occurred before the barge drifted away and the towline, which was fouling the propellers, parted. With *Shearwater* anchored

and powerless there was no plan or safe course of action for the crew to recover the situation; the nature of this emergency then required an extensive effort by a lifeboat, an ETV and a workboat to be recovered.

The hazards associated with this operation had not been assessed or recognised and, therefore, there was no safe system of work to control the risks. This happened primarily due to the lack of crew experience with coastal towage and their inability to recognise and mitigate such risks. After a long, tiring day, and in an effort to seek shelter, the unplanned and unsupervised confusion with ad-hoc procedures meant that the fouling of the propellers and subsequent collision were effectively inevitable.

2.4 TOWING

2.4.1 Tow planning

There are many potential hazards involved in towing a laden barge, including the potential for towline failure, capsize of the barge or even capsize of the towing vessel. To manage these risks, coastal towing requires suitable equipment, skilled crew, and a robust plan. The tow plan should include details of the towing equipment, tow route and emergency plans.

Although *Shearwater*'s owner had prepared a tow plan (**Annex C**), this was not based on any recognised planning framework such as the IMO's Circular or the WA guidance. The tow plan indicated that the alongside tow was to be used in '*sheltered conditions*'. Although the initial part of the voyage on the day of the accident was sheltered, the area around Cape Wrath and beyond was exposed to the risk of ocean swells. It was not appropriate to attempt to tow *Agem One* alongside in these areas, as evidenced by the difficulties the crew actually encountered.

Furthermore, the tow plan did not adequately address the tow set up; at 80m long, the rope in use was far too short given industry guidance that 500m was a practical minimum for coastal towing. Additionally, the plan referred to Kevlar and polyester towlines when, in fact, an HMPE line was in use. Deployment of a short HMPE rope without a 'stretcher' made the rig very vulnerable to snatch loading, risking failure.

The plan also did not contain procedures to be followed in an emergency, which resulted in extremely hazardous personnel transfers in deteriorating conditions to recover the drifting barge. Additionally, although the plan discussed snap back and crush zones, these were not defined and there was no guidance or deck markings on either vessel to show where these hazards were.

Not following recognised guidance meant that the tow plan was wholly inadequate for the towing operation being considered. This happened primarily because of the owner's lack of familiarity with coastal towing and his ambition to press ahead with the voyage. Equally, the master did not challenge the towing plan; however, the owner was on board making all the key decisions and had effectively discharged the master of any responsibility for towing by the employment contract amendment [Section 1.6.2].

2.4.2 Towline failure

Although the initial failure of the towline near Cape Wrath did not directly contribute to the accident, it started the chain of events that led to the collisions. The cause of the towline parting has not been determined and the failed end of the rope was not recovered from *Agem One*, so it was not inspected or assessed. Nevertheless, given the very strong HMPE rope in use, a tension overload can be discounted. The reported distortion of the hard eye suggests that the towing bridle and shackle may have been incompatible potentially generating a twisting movement that distorted the eye. Equally, the absence of a stretcher or sufficient length to generate a catenary, meant the towline would have been susceptible to snatch loading. With the hard eye reported as distorted, the towline would have been vulnerable to chafe abrasion caused by the barge's bridle shackle.

The reported subsequent failures, including after *Shearwater* had anchored, were almost certainly due to abrasion when the HMPE rope had been tied in a knot, and then severely damaged by the propellers.

Although the towline failures did not directly contribute to the accident, they can be attributed to inappropriate practices including insufficient tow length and onboard securing arrangements that introduced a risk of snatch and chafe.

2.4.3 Suitability of *Shearwater* for coastal towing

Purpose-built tugs are highly manoeuvrable, fitted with dedicated towing equipment and have sufficient power to tow at adequate speeds to maintain forward progress and manoeuvre out of danger if required. The charter contract between *Agem One*'s owner and *Shearwater*'s owner required that a tug of a 'suitable size' should be used when it was towed.

When towing *Agem One*, *Shearwater* could only maintain a speed of between 3 and 5.2kts. At these low speeds, *Shearwater*'s manoeuvrability was reduced and it would have been very challenging to make headway in any adverse wind and tide. Without a winch that was certified as suitable for towing, there was also no practical method of adjusting the towline's length. Monitoring the tow would also be very difficult as *Shearwater*'s bridge did not have purpose-built visibility aft.

Shearwater was designed and built as a small harbour dredger and lacked the power or towing equipment to undertake a coastal towing operation, especially given the hazards associated with the Pentland Firth.

2.5 EMERGENCY RESPONSE

2.5.1 Loss of control of the barge

Although *Agem One* was fitted with an emergency towline, this had not been prepared or streamed astern which meant that there was not a safe method for recovering the barge when the main towing arrangement failed. This occurred on at least two occasions and, without an emergency towline, the crew had to undertake an ad-hoc procedure involving unsafe personnel transfers to recover the situation. When the towline parted after the propeller shafts were fouled, *Agem One* was drifting free as a hazard to other shipping and a very significant task for the lifeboat to recover.

2.5.2 Bilge pumping

It is vital for the safe operation of any vessel that floodwater, which creates an immediate hazard, should be reliably pumped out using all available means. *Shearwater's* emergency pumping procedure stipulated the use of the general service pumps in the first instance to counteract flooding. Despite the crew attempting to use this system it could not be made to function. The flooded space had a bilge suction and there were no defects with the system. However, the crew lacked the familiarity necessary to make the system work effectively in an emergency. The main bilge system had not been lined up to pump from the port buoyancy space for 8 years. This was exacerbated by the absence of a dedicated engineer, or a crew member formally delegated responsibility for machinery systems, and the diagram that dated from 1968 not accurately reflecting the onboard system. Furthermore, once the portable electric pump started to take effect, this was relied upon for dealing with the emergency. The petrol driven salvage pump could not be used having been damaged by the water ingress.

2.5.3 Raising the alarm

Shearwater's owner only raised the alarm when *Agem One* broke away for the final time, because of a concern that the unlit barge represented a hazard to other vessels. However, prior to this, the crew had been struggling for many hours with towline failures, culminating in immobilisation and flooding.

On board *Shearwater*, there was an illusion of control, where the owner persistently believed that the situation could be resolved without external assistance. However, the onboard flooding checklist included a reference to raising the alarm, which underpins the analysis of the absence of control, and that the procedure itself was not being followed.

2.6 CREW COMPETENCE

Coastal towing is a skilled job, requiring suitably experienced and competent crew and *Agem One's* certification required tow preparations to be to the '*satisfaction of the tow master*'.

In the absence of a formal definition of what constituted a tow master, and in line with the WA towing guidance, the MCA's voluntary towage endorsement scheme provided a suitable guide for the experience necessary. *Shearwater's* owner had no formal training in towing operations and his towing experience was less than the suggested 120 days required to meet the standard for general towage under the MCA's scheme, let alone the 300 days required for sea towage. Additionally, the majority of the owner's towing experience was restricted to harbour towing. *Shearwater's* master had no prior towing experience using a dredger and was wholly reliant on the owner to manage the towing operations. Thus, there was no-one onboard suitably experienced to act as the tow master and effectively supervise the towing operation.

Furthermore, the MCA had, in their discussions and subsequent conversations with *Shearwater's* owner in 2014, identified that crew competence for towing was an issue that needed to be addressed if the vessel was to be used more regularly for towing. However, this was not followed up by either the owner or the MCA. Despite his experience on the vessel, *Shearwater's* owner was not a qualified professional

seafarer. Although he was on board, the owner had employed professional crew to operate the vessel for him, but they were not suitably qualified and experienced for the intended operations. Equally, the MCA placed no obligation on the owner to ensure that a suitable crew were manning the vessel, despite the potential that *Shearwater* was likely to be used for towing.

Even where a suitably qualified and experienced tow master is on board, the master has overall responsibility for the vessel and should make the final decisions on passage planning and safe operations. Although effectively discharged by the owner from any responsibility for towing due to his lack of experience, the master still had a role to plan and execute other aspects of the vessel's operations and management, including safe navigation. However, an awkward situation of divided responsibility between the owner and the master existed, with an untidy lack of clarity over who was in overall control. Ultimately this undermined the master's position and created an unhelpful and potentially unsafe absence of authority.

2.7 PASSAGE PLANNING

Effective passage planning requires consideration of a broad range of documentary evidence and the preparation of a comprehensive and safe plan. Effective passage plans are also dynamic in their execution, taking variable elements such as weather and tidal streams into account as the voyage progresses.

Shearwater's written passage plan was insufficiently detailed and lacked any information about the potential navigational hazards. The route plotted on the paper charts and the ECS (**Figure 9**) was an overall construct of the complete voyage and not berth to berth passage plans for each day. This was not a plan that would safely underpin critical safety-related decision making during the passage.

Evidence from AIS suggests that *Shearwater* towing *Agem One* was passing at between 3kts and 5.2kts. These are speeds where the tidal stream is critical to the plan and, in effect, any predicted stream of more than about 1kt would be significant. With strong tidal streams predicted in the Pentland Firth (information that could have been identified on board from the chart), the passage ahead would have appeared unachievable when passing Cape Wrath. Therefore, it was probably most fortunate for the crew, that the swell and the towline failure forced abandonment of the voyage before reaching the challenging conditions local to the Pentland Firth.

Closure of the Caledonian Canal and the inability to offload the excavator in Lochaline, both a result of COVID-19, introduced significant new risks to the plan, specifically the need to pass through the potentially treacherous Pentland Firth. Both these events demanded a formal reappraisal of the plan and consideration could have been given to cancelling the voyage altogether for safety reasons or chartering a tug to relocate the barge instead of using *Shearwater* for towing. However, the owner's decision was to press ahead with the voyage in order to undertake the dredging contract in Eyemouth. Having made significant progress with the repositioning voyage, the owner was likely to be motivated to continue to the destination and deliver the forthcoming contract. Equally, the decision to press ahead with the voyage in the face of emergent new hazards went unchallenged by the master, who had a critical role of planning and executing the passage. This almost certainly occurred because the master lacked experience of the industry sector and the local hazards, and the owner was on board and making all the key decisions. Furthermore, the nature of Northern Dredging's organisation meant there was also no external governance of the vessel's operations.

Shearwater's passage plan was over-ambitious and unsafe. The plan itself did not address or mitigate the foreseeable navigational hazards and insufficient consideration was given to amending or cancelling the passage as the situation evolved. The owner was driven by a desire to complete the voyage in order to commence the next contract and the master did not sufficiently challenge the owner's decision making, or deliver a safe plan.

2.8 SAFE MANNING

The MCA's requirement for the safe manning of *Shearwater* was unclear. The load line exemption certificate required *Shearwater* to be manned in accordance with the requirements for a Class VIII(A) vessel and also stated that there should be a crew of four. However, there was no elaboration as to what these statements meant in practice, or the level of qualifications required by each member of the crew. Moreover, working at sea must be guided by the MLC requirement to provide a minimum of 10 hours rest in any 24-hour period, necessitating a minimum of two watchkeepers whenever a vessel is expected to be at sea for periods where the minimum rest requirements of a lone watchkeeper are not achievable.

In the 8 years that the owner had operated *Shearwater*, its manning had been discussed internally by the MCA several times, and concerns had been raised about the vessel operating with a single officer of the watch. Other than a short-lived restriction on *Shearwater*'s certificate limiting voyages to 12 hours duration, these concerns had not been dealt with and the required safe manning arrangements were never clarified or included in MCA documentation.

Shearwater's owner had, prior to this accident, recognised that a second bridge watchkeeper was required when operating in areas with long transits to the spoil grounds, but he had not employed a mate for this repositioning voyage. However, evidence from AIS indicated that passages exceeding the MLC requirement were undertaken with only one watchkeeper on board. Campbeltown to Oban is 99nm and would have taken between 21 and 29 hours based on typical speeds being achieved by *Shearwater* towing *Agem One* at the time. This was a period of time underway that exceed the MLC requirement where only one suitably qualified watchkeeper was on board. With only a stop of a few hours at Stoerhead, *Shearwater* had also effectively been at sea for about 36 hours when the first break occurred in the towline.

Without detailed guidance, *Shearwater*'s owner was able to crew the vessel as he saw fit, rather than in accordance with a clearly defined code. Scant regard was paid on board to the minimum rest requirement of the MLC, introducing a severe risk of crew fatigue. Having made good progress, the owner was motivated to sustain the passage in the face of adversity, such as not being able to offload the excavator, and irrespective of the hazards ahead.

2.9 SAFETY MANAGEMENT

The safe operation of a vessel is delivered through effective safety management by identifying hazards, assessing risks and implementing safe systems of work.

Underway personnel transfer, with the associated hazards of crushing or man overboard, is a good example of an operation requiring an assessment of the risk and a safe system of work. Industry guidance also existed for this operation, which suggested that it should only be conducted when absolutely necessary.

Changing the tow configuration during the voyage and setting up the tow after towline failures required the transfer of crew between *Shearwater* and *Agem One* when the dredger was underway. This was an unsafe practice that occurred because the risks had not been assessed and no safe system of work existed. Without a formal requirement for a safety management system and only a remark on the load line exemption certificate requiring the vessel to be managed '*ensuring the safety of personnel*', there was insufficient guidance or supervision of safe operations on board *Shearwater*.

2.10 FLAG STATE OVERSIGHT

In November 2020, *Shearwater* was one of 512 vessels on the UK register that fell between the small vessel codes of practice and the SOLAS regulations. The potential benefits of a consolidated code of practice for vessels above 24m load line length and below 500gt were highlighted in the MAIB's report in 2009 after the loss of *Abigail H*. In collaboration with industry partners, the MCA developed such a draft code of practice; however, this was paused in 2018 due to the lack of a pathway to develop the underpinning legislation necessary for an enforceable code. This meant that the requirements for manning and safety management for *Shearwater* were linked to conditions on the dredger's load line exemption certificate, despite these areas not falling within the scope of the Load Line Regulations.

Load line exemption certificates set out those parts of the regulations that are suspended for the vessel in question; however, *Shearwater's* certificate exempted the vessel from all requirements of the Load Line Regulations, including requirements for annual survey and minimum freeboard, and contained no reference to dredging or towing. There were also significant inconsistencies in *Shearwater's* previous certificates with some making reference to the vessel's operation as a dredger and others not. Disapplying all requirements of the Load Line Regulations was not appropriate for an 18-month certificate for a vessel of *Shearwater's* age and history.

In 2014, *Shearwater's* owner approached the MCA to discuss undertaking the coastal towage of a barge. As a result of this discussion, the owner commissioned a set of strength calculations for the bollards but there was no follow up by either party to finalise the issue and determine the towing conditions. Although there is a section on towing in the MCA surveyor's aide-memoire, this was only applied on surveys of towed objects; there was no prompt for surveyors to consider vessels used for towing. As *Agem One* was certified in its own right, no load line exemption, with consequent consideration of the towing vessel, was required. Equally, *Shearwater's* load line exemption certificate made no reference to the use of *Shearwater* for towing at sea.

Shearwater was well known to MCA surveyors, having been previously detained, and, in the 8 years since purchase, had been issued with 18 certificates. Throughout this time there were several discussions and extensive correspondence albeit with no documented resolution between the owner and MCA surveyors about the operation of the vessel, including appropriate manning and suitability for towing. Unfortunately, the owner had not demonstrated a strong commitment to the MCA's governance, evidenced by repeated detentions for unsafe operations. Nevertheless, at the most recent survey in 2019, the MCA had noted his improved engagement with the survey process.

In summary, despite the MCA's extensive and detailed engagement with *Shearwater's* owner, the issues of manning, towing and safety management were left unresolved and did not result in enhanced vessel requirements or additional conditions on *Shearwater's* load line exemption certificate. The MCA's surveyors invested time and energy on *Shearwater's* technical details; however, the wider picture of safe operations were not effectively addressed. *Shearwater's* load line exemption certificate did not provide sufficient guidance to assure safe operations and created an environment where the owner felt able to operate the dredger broadly as he saw fit, rather than safely in accordance with a recognised code of practice.

SECTION 3 – CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. *Shearwater*'s hull was breached, and the vessel flooded after several heavy collisions with the towed barge *Agem One*. The collisions occurred because control of the barge was lost when *Shearwater* was immobilised. [2.3]
2. *Shearwater* was immobilised when astern propulsion was applied with the towline was in the water. This happened because the crew did not have a safe system of work for connecting the tow. [2.3]
3. *Shearwater*'s towing operations were not effectively planned or safely executed. This happened primarily because industry best practice was not followed, and the crew lacked the competence necessary to undertake an operation of this nature or fulfil the role of tow master. [2.4.1, 2.6]
4. *Shearwater*'s crew did not respond effectively to the emergency. They were not properly prepared to deal with the towline failure, and were unable to use the fixed bilge system effectively due to a lack of familiarity with this critical safety system. [2.5]
5. *Shearwater*'s passage plan was over-ambitious, unsafe and did not effectively address the potential hazards. In particular, the owner intended to press ahead with a potentially untenable voyage through the Pentland Firth after the situation had changed. The owner was motivated to continue the voyage in the face of adversity, despite the emerging risks. [2.7]
6. *Shearwater* was not safely manned for the intended voyage, and evidence from AIS indicates that the vessel was underway for unacceptably long periods with only one bridge watchkeeper on board, risking fatigue. [2.8]
7. *Shearwater*'s master and owner did not have a robust process of identifying hazards, assessing risks and developing safe systems of work. This happened primarily because there was no requirement for a safety management system. [2.9]
8. The MCA's load line exemption certificate did not provide sufficient guidance to assure safe operation of the vessel. Exempting *Shearwater* from all conditions of the Load Line Regulations was inappropriate, and the certificate's conditions were insufficiently detailed with respect to manning and safety management. Additionally, the certificate did not set conditions for *Shearwater*'s operation as a dredger or towing vessel. [2.10]
9. The absence of specific flag state guidance on the safe operation of the vessel created an environment where the owner felt able to operate the vessel broadly as he saw fit, rather than safely in accordance with a recognised code of practice. [2.10]

3.2 SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. Without sufficient power or towing equipment, *Shearwater* was not suitable for use as a coastal towing vessel, especially given the potentially severe conditions likely to be encountered in the Pentland Firth. [2.4.3]
2. The towline failures resulted from inappropriate towing practices that introduced a significant risk of snatch and chafe. [2.4.2]

SECTION 4 – ACTION TAKEN

4.1 ACTIONS TAKEN BY OTHER ORGANISATIONS

The **owner of *Shearwater*** has:

- Purchased a small tug for use when repositioning barges, which means that *Shearwater* is no longer required to be used for coastal towing.
- Updated the onboard documentation for the bilge system.
- Introduced further familiarisation training of machinery systems for all crew and new joiners.
- Introduced a requirement for the master to operate bilge pumping systems and propulsion machinery.
- Commenced the process of obtaining a voluntary towage endorsement for one master to operate the small tug.